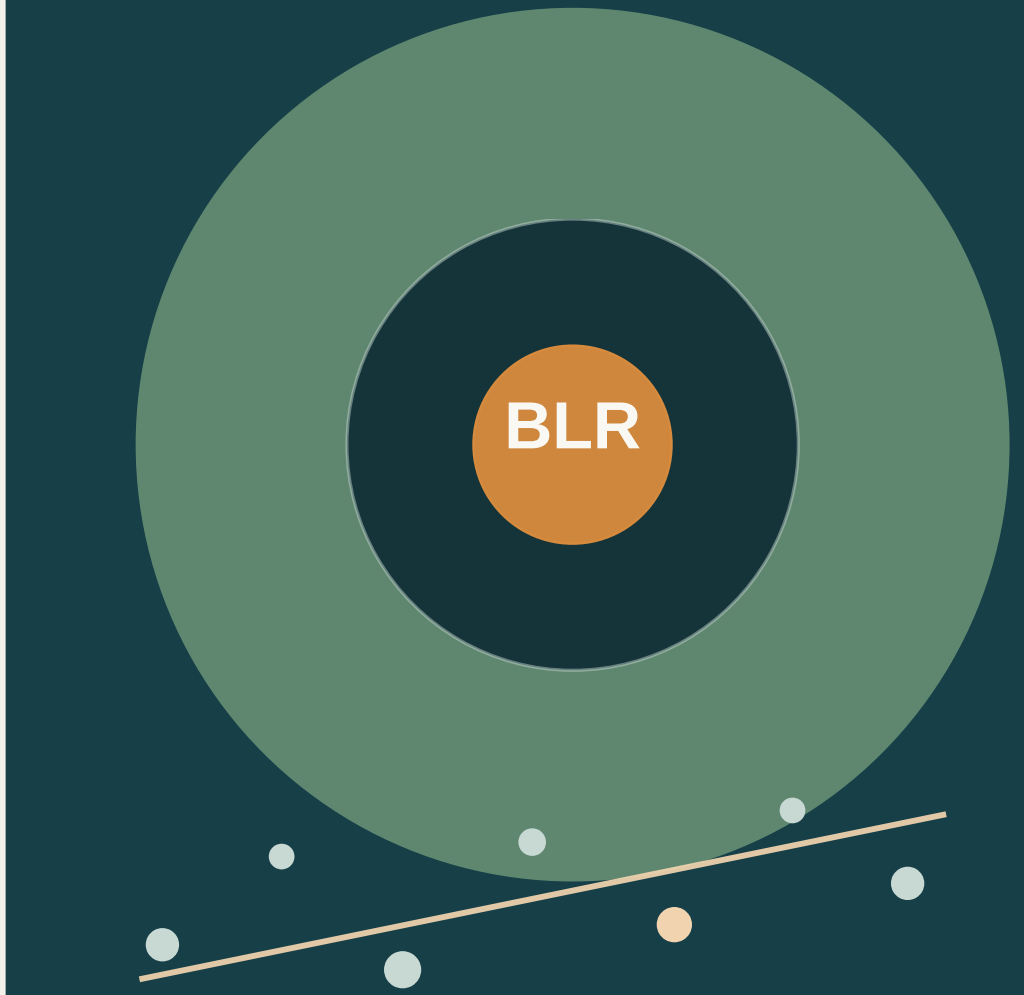
BEYOND MULTIPLE LINEAR REGRESSION

Randomized block designs

Modeling center effects and comparing treatments

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METHODS · DESIGN · STATISTICS

Why go beyond multiple linear regression?

- Observations often share a setting or unit, such as patients within a center or repeated measurements from one person.
- Differences between groups contribute to variation in outcomes, alongside differences between individuals within groups.
- Ignoring this structure can bias effect estimates or give incorrect standard errors, affecting confidence intervals and hypothesis tests.
- In this course, you learn to choose, fit and interpret models that account for grouping, separating variation between groups from variation within groups.



How the course works

A weekly cycle of preparation, guided practice and independent application.

WHEN	WHAT YOU DO
Before class	Read the selected book chapters to prepare for the session.
During class	Build understanding through lectures and develop data analysis skills in guided R labs.
After class	Apply the readings and in-class material to a new dataset in the weekly assignment.

Assignments and reproducible reporting

Use Quarto to bring your analysis and interpretation together.

ONE REPRODUCIBLE REPORT

A Quarto source file (.qmd) combines written explanations with R code.

Rendering runs the analysis and produces a report containing the results.

Explain your choices and interpret the findings alongside the analysis.

WHAT TO SUBMIT

The Quarto source file (.qmd), including all R code and annotations.

The rendered HTML report, with the R code visible.

Use the provided assignment template.

Submit both the source and the rendered report so the analysis can be read and reproduced.

Certificate and assessment

COURSE CERTIFICATE

Participate in at least 80% of the in-class meetings.

Submit the weekly data analysis assignments.

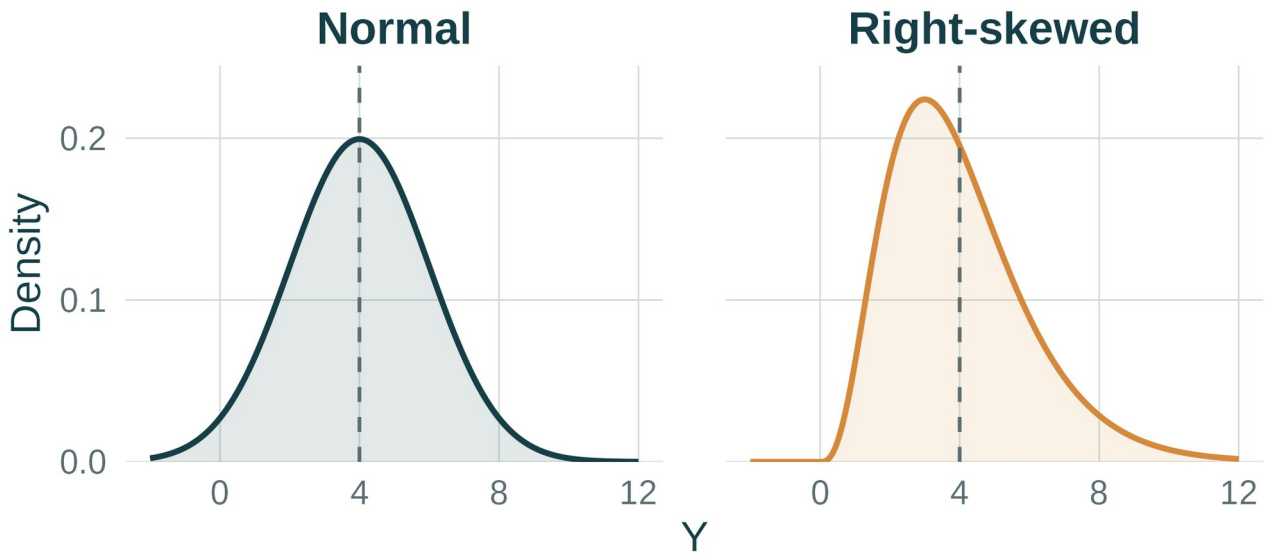
IF YOU REQUIRE A GRADE

Your grade is based on the final data analysis assignment.

This applies, for example, to research master students and PhD students seeking Epidemiologist B registration.

Expected value, variance and standard deviation

- Y is a random variable: its value can differ across repeated realizations.
- Expected value μ is its long-run average, or balance point. It need not be the peak.
- Variance σ² is the average squared deviation from μ.
- Standard deviation σ is the square root of the variance, in the same units as Y.



Both: mean 4, variance 4, SD 2. Dashed lines mark the means.

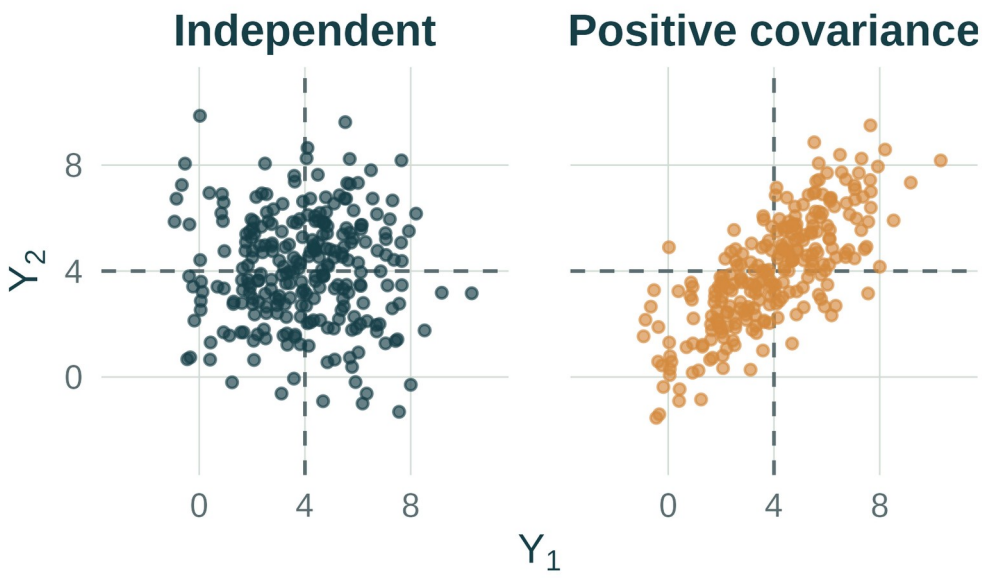
$$E(Y) = \mu$$

$$\mathrm{Var}(Y) = E\left[(Y-\mu)^2\right] = \sigma^2$$

$$\mathrm{SD}(Y) = \sqrt{\mathrm{Var}(Y)} = \sigma$$

Covariance and independence

- Each point is a pair of values of Y_1 and Y_2.
- Covariance measures how deviations from the two means vary together.
- Independence means knowing one variable does not change the distribution of the other.
- Independence implies zero covariance. The converse does not generally hold.



Both variables have mean 4 and variance 4.
Dashed lines mark the expected values.

$$\mathrm{Cov}(Y_1, Y_2) = E[(Y_1 - \mu_1)(Y_2 - \mu_2)]$$

Positive covariance: deviations from the two means tend to have the same sign.

Linear combinations of random variables

- A linear combination multiplies random variables by constants and adds them.
- Expected values combine using the same weights.
- Variance depends on the squared weights and on covariance.
- For independent variables, the covariance term is zero.
- Adding a constant changes the mean, but not the variance.

$$Z = a + bY_1 + cY_2$$

a, b and c are constants; means and variances are finite.

EXPECTED VALUE

$$E(Z) = a + bE(Y_1) + cE(Y_2)$$

VARIANCE

$$\operatorname{Var}(Z) = b^2 \operatorname{Var}(Y_1) + c^2 \operatorname{Var}(Y_2) + 2bc \operatorname{Cov}(Y_1, Y_2)$$

The expectation and variance rules apply regardless of distribution.
If Y_1 and Y_2 are independent and Normally distributed, Z is also Normally distributed.

Useful special cases

Write $\mu = E(Y)$ and $\sigma^2 = \text{Var}(Y)$, with subscripts for Y_1 and Y_2.

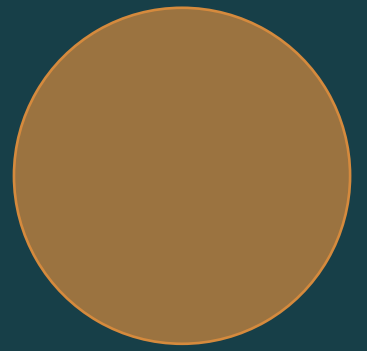
TRANSFORMATION	RANDOM VARIABLE	EXPECTED VALUE	VARIANCE
Add a constant	$a + Y$	$a + \mu$	σ^2
Rescale	bY	$b\mu$	$b^2\sigma^2$
Sum (independent)	$Y_1 + Y_2$	$\mu_1 + \mu_2$	$\sigma_1^2 + \sigma_2^2$
Difference (independent)	$Y_1 - Y_2$	$\mu_1 - \mu_2$	$\sigma_1^2 + \sigma_2^2$

For independent variables, both a sum and a difference have variance $\sigma_1^2 + \sigma_2^2$.

PART 1

A multicenter trial: initial analysis





A multicenter rehabilitation trial

DESIGN AND OUTCOME

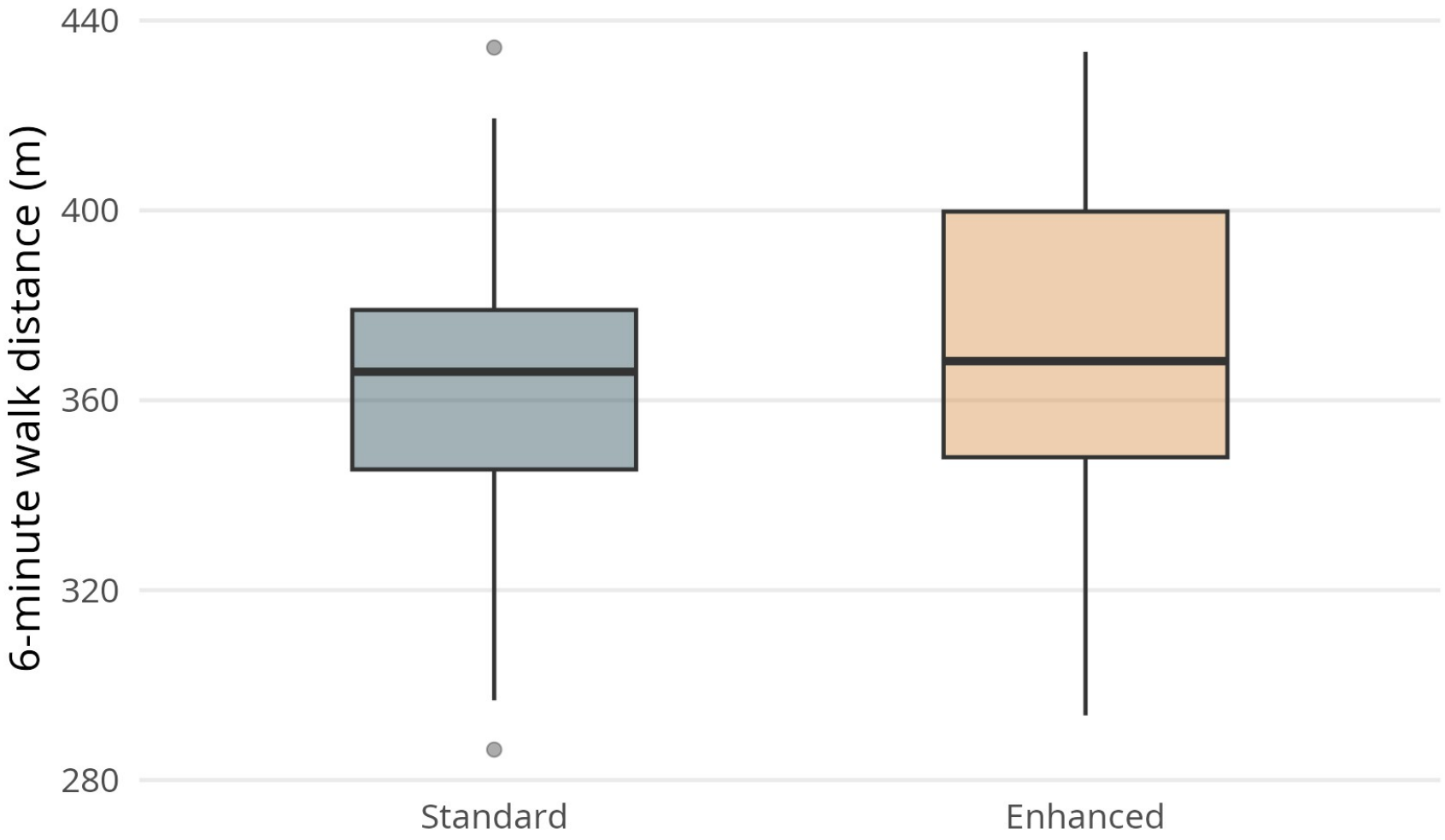
- Simulated trial: 120 patients after knee replacement at 6 centers.
- Randomized block design: centers are the blocks. Within each center, 10 patients are randomized to each treatment.
- Standard care: weekly physiotherapy and home exercises.
- Enhanced care: standard care plus a weekly supervised session.
- Outcome: six-minute walk distance at 12 weeks.

RESEARCH QUESTION

How much do the mean walking distances differ between the programs?



Walking distances pooled across centers



60 patients per program across all 6 centers.

The treatment-only model

Patient i, center j: Y_{ij} is walking distance; T_{ij} = 0 (standard) or 1 (enhanced).

SYSTEMATIC PART AND ERROR

$$Y_{ij} = \underbrace{\beta_0 + \beta_1 T_{ij}}_{\text{model-predicted outcome}} + \underbrace{u_{ij}}_{\text{individual error}}$$

β_0 is the expected walking distance under standard care.

β_1 is the expected difference between enhanced and standard care.

u_{ij} is an individual's deviation from the model-predicted outcome.

ASSUMPTIONS AND IMPLICATIONS

$$u_{ij} \overset{\text{iid}}{\sim} N(0, \sigma_u^2)$$

Independent Normal errors, with mean zero and a common variance.

For a given treatment, $\beta_0 + \beta_1 T_{ij}$ is a constant.

Applying the recap rules: adding a constant (slide 9)

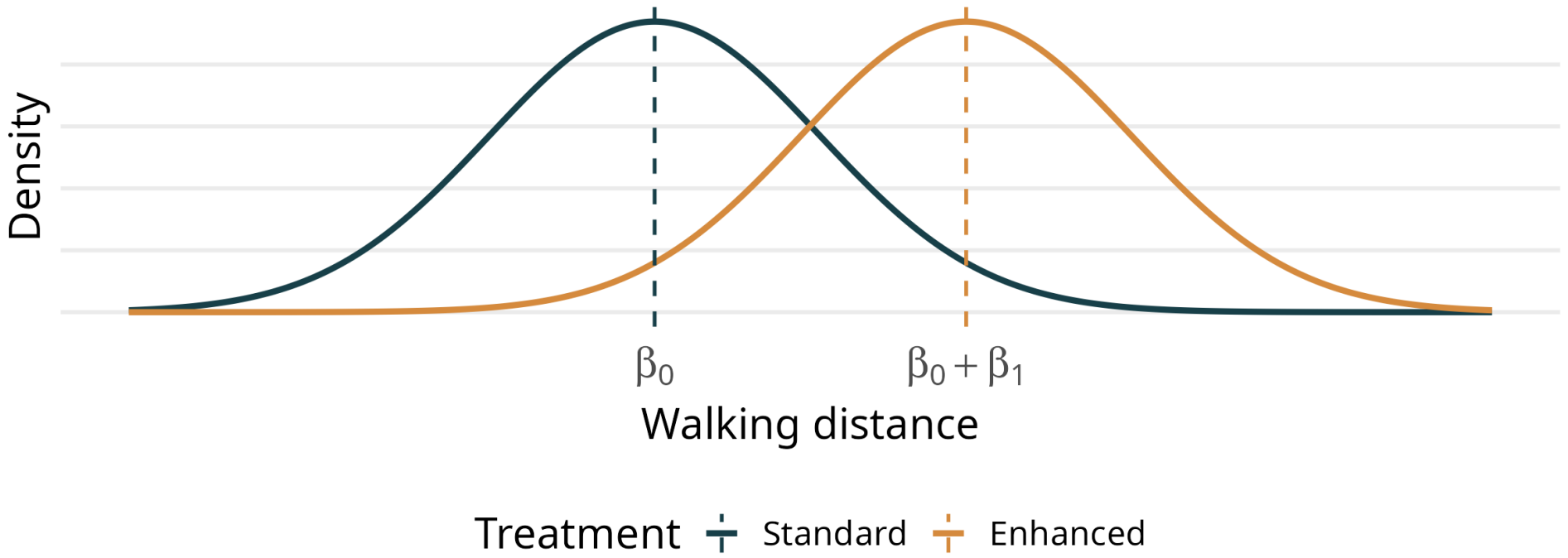
$$\begin{aligned} E(Y_{ij}) &= \beta_0 + \beta_1 T_{ij} + E(u_{ij}) \\ &= \beta_0 + \beta_1 T_{ij} \end{aligned}$$

$$\text{Var}(Y_{ij}) = \text{Var}(u_{ij}) = \sigma_u^2$$

From errors to outcome distributions

ADDING A CONSTANT PRESERVES THE NORMAL SHAPE

$$Y_{ij} \sim N\left(\beta_0 + \beta_1 T_{ij},\ \sigma_u^2\right)$$



The means differ by β_1. Both treatment groups have the same spread: SD σ_u.

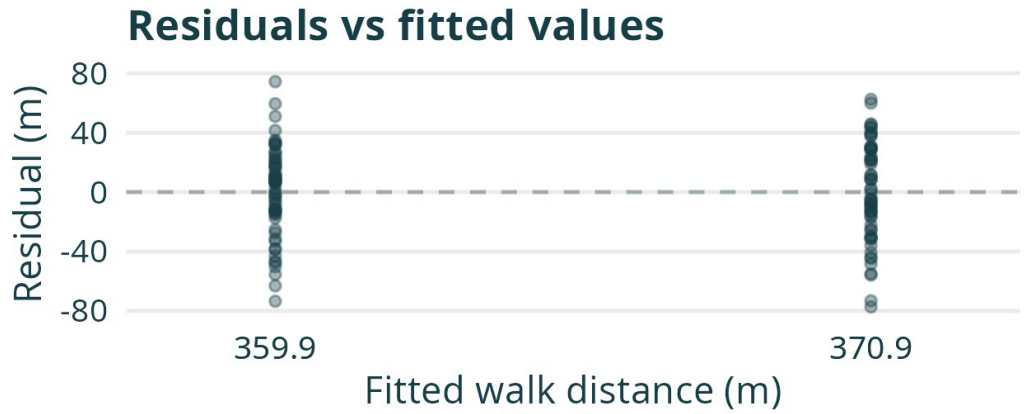
Model results

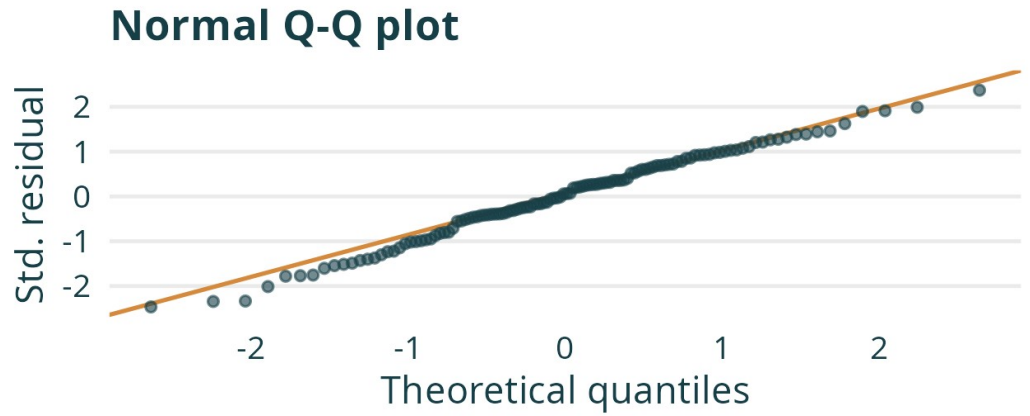
Term	Estimate (m)	SE (m)	95% CI (m)	p-value
Intercept (β_0)	359.9	4.08	351.8 to 368.0	< .001
Enhanced care (β_1)	11.0	5.77	-0.4 to 22.5	0.059

Pooling across centers, patients assigned to enhanced care walked 11.0 m farther on average.

Estimated residual variance: 1000.2 m².
Corresponding residual SD: 31.6 m.

Model diagnostics





WHAT WE USE THIS PLOT FOR

Assess whether residuals are centered around zero and have similar spread across fitted values.

WHAT WE USE THIS PLOT FOR

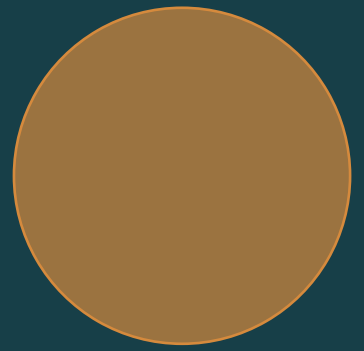
Assess whether the residual distribution is approximately Normal.

The plots suggest no clear violations of Normality or equal variance.
They do not establish whether ignoring center gives valid inference.

PART 2

Modeling center as a fixed effect





A multicenter trial, a pooled analysis

- The trial used a randomized block design: patients were randomized to the two programs within each of six centers, with equal numbers per program in every center.
- The treatment-only model represents each treatment group as one homogeneous sample, without distinguishing the centers from which patients were recruited.
- Centers may differ in patient populations and care settings, shifting expected outcomes for both treatments.

Do center differences in expected outcomes matter for our treatment comparison?



Omitting center: a mathematical explanation

THE DATA-GENERATING MODEL

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + c_j + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

c_j is center j's fixed shift from the average expected outcome. The center shifts sum to zero.

After accounting for treatment and center, individual errors are independent Normal variables with mean 0 and variance σ^2.

THE TRUE ERRORS WHEN CENTER IS OMITTED

We can rewrite the same generated outcomes as:

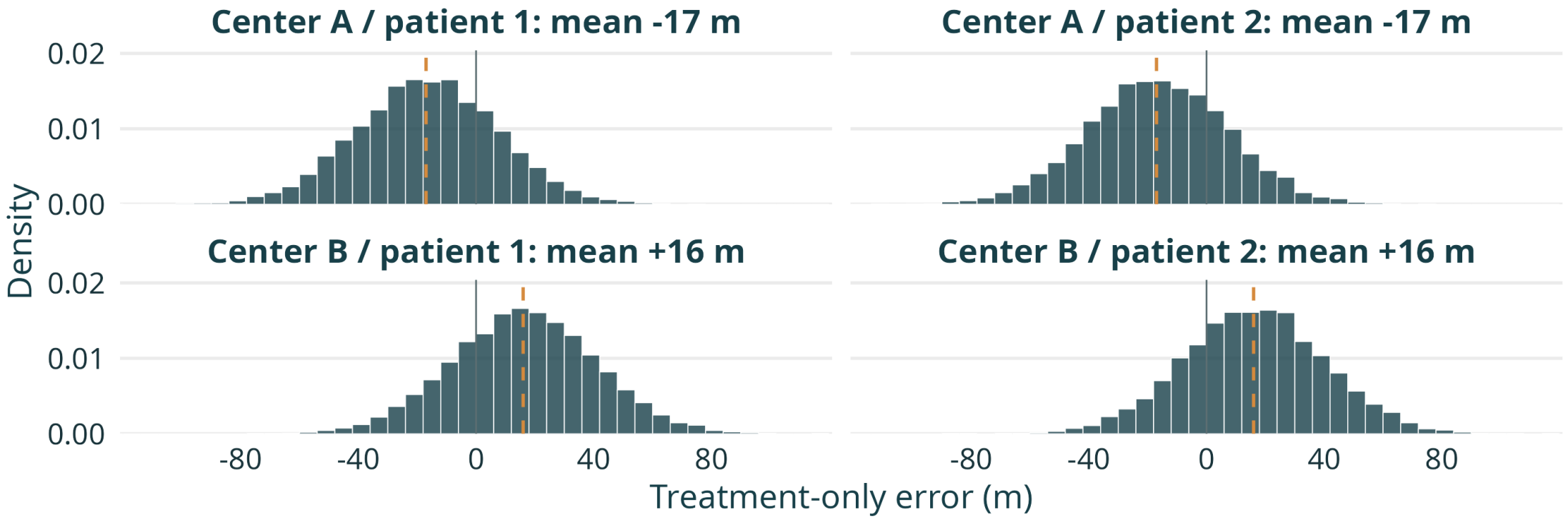
$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_{ij}$$

$$\boxed{u_{ij} = c_j + \varepsilon_{ij}}$$

$$u_{ij} \sim N(c_j, \sigma^2)$$

These errors remain independent, with equal variances, but their means differ between centers.

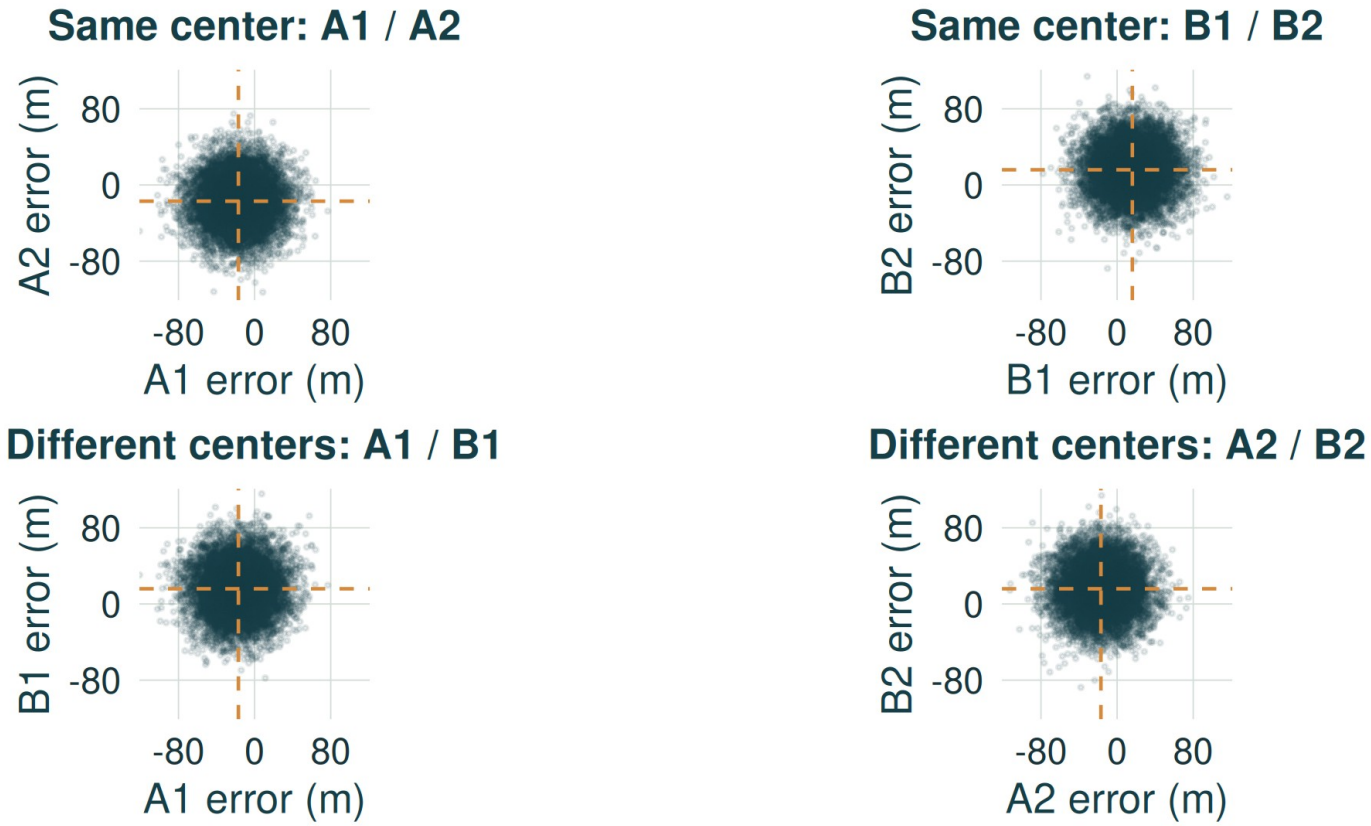
Errors: different means, same variance



10,000 studies with fixed center effects; 10 patients per treatment in each center; individual-error SD = 24 m. Center effects (m): A −17, B +16, C +8, D −29, E +27, F −5. Dashed lines: expected errors.

The means differ by center; the variation around each patient's own mean is the same.

Errors: independent within and between centers



Each dot: two errors from one of 10,000 studies with fixed center effects. Dashed lines: expected errors.

A fixed center effect shifts the means without creating dependence.
Theoretical covariance = 0; empirical covariances fluctuate around zero.

True errors and the treatment-only model

THE TRUE ERROR DISTRIBUTION

$$\boxed{u_{ij} = c_j + \varepsilon_{ij}}$$

$$u_{ij} \sim N(c_j, \sigma^2)$$

Independent errors.

Means differ between centers: c_j.

Equal variances: σ^2.

WHAT THE FITTED MODEL ASSUMES

$$u_{ij} \overset{\text{iid}}{\sim} N(0, \sigma_u^2)$$

A common zero-mean Normal distribution.

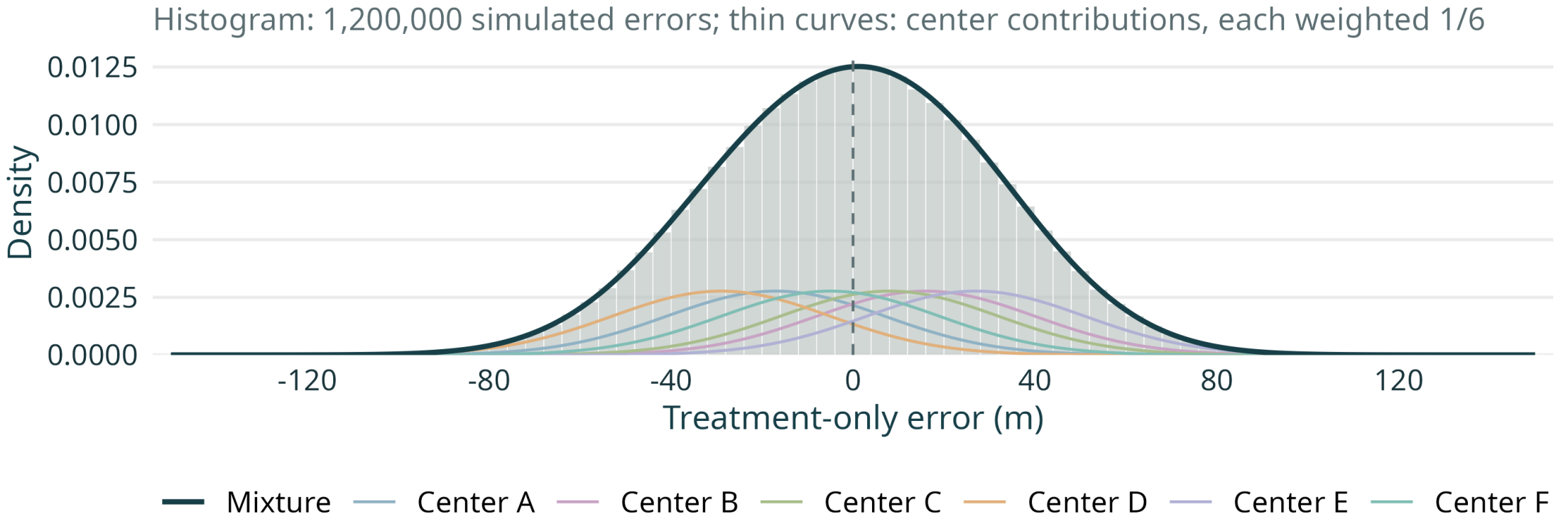
Independent errors.

Mean zero for every patient.

One common variance, σ_u^2, estimated from the data.

The treatment-only model correctly assumes independence and equal variances, but misses the center-specific error means.

The pooled error distribution



Each center contributes equally; the center effects remain fixed across simulations.

The pooled error distribution is a mixture of six center-specific distributions. Because the centers contribute equally and their effects average to zero, the mixture also has mean zero.

Balanced allocation keeps the treatment estimate unbiased

BALANCE MAKES CENTER DIFFERENCES CANCEL

$$\widehat{\beta}_1 = \bar{Y}_{\text{enhanced}} - \bar{Y}_{\text{standard}}$$

Both groups contain 10 patients from every center. The shared baseline and average center contribution cancel.

$$\widehat{\beta}_1 = \beta_1 + \bar{\varepsilon}_{\text{enhanced}} - \bar{\varepsilon}_{\text{standard}}$$

Each epsilon bar averages the individual errors in that treatment group.

These individual errors have mean zero, so:

$$E(\widehat{\beta}_1) = \beta_1$$

Balance between the treatment groups makes center differences cancel from the treatment comparison.

Omitting center inflates the estimated standard error

WHAT DETERMINES THE TRUE SE?

Center differences cancel from the treatment contrast.

Only individual errors contribute to its sampling variability.

$$\mathrm{SE}(\widehat{\beta}_1) = \sigma\sqrt{\frac{1}{60} + \frac{1}{60}}$$

σ is the individual-error SD: variation around the center-specific means.

THE POOLED SPREAD ENTERS THE ESTIMATED SE

Pooled errors include variation within centers and differences between center means.

One fitted mean per treatment group leaves center differences in the residuals.

$$\widehat{\mathrm{SE}}(\widehat{\beta}_1) = \widehat{\sigma}_u\sqrt{\frac{1}{60} + \frac{1}{60}}$$

Residual SD reflects the pooled spread rather than the individual-error SD σ.

Center differences do not contribute to the treatment contrast's variability,
but they inflate the residual SD used to estimate its SE.

Overestimated standard errors overstate uncertainty

CONFIDENCE INTERVALS

A confidence interval uses the estimated SE to express uncertainty around the treatment estimate.

An inflated SE makes the confidence interval unnecessarily wide.

The reported uncertainty is greater than the actual sampling uncertainty.

HYPOTHESIS TESTS

A test compares the treatment estimate with its estimated SE.

An inflated SE makes the absolute test statistic smaller, generally giving a larger p-value.

We have less power to detect a real treatment effect.

Accounting for center can give narrower, appropriately calibrated confidence intervals and greater power, without changing the treatment estimate in this balanced design.

Time for lab 1



Lab 1 debrief: accounting for center

WHAT CHANGES IN THE MODEL

- Including center allows the fitted outcomes to reflect systematic differences between centers.
- Those differences no longer remain in the residuals.
- The residual SD now estimates the individual-error SD:

$$\widehat{\sigma} = 23.8 \text{ m}$$

WHAT CHANGES IN THE SE

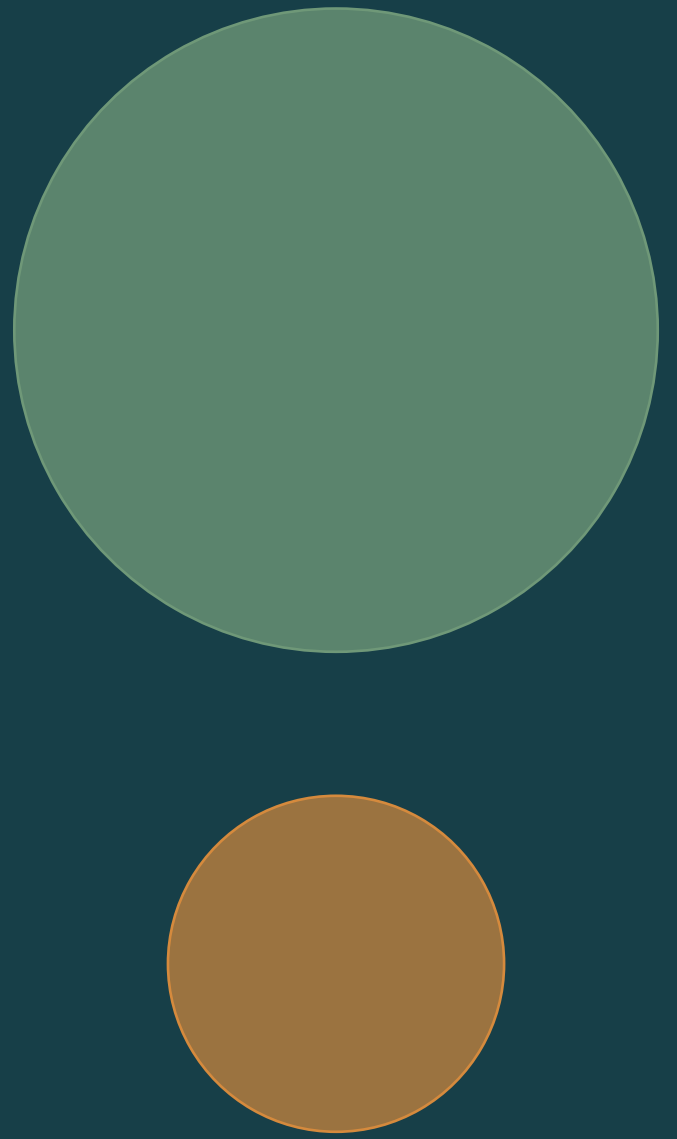
- The treatment estimate remains 11.0 m, with the same sampling distribution: balanced allocation already makes center contributions cancel.
- The SE now uses the estimated individual-error SD:

$$\widehat{\mathrm{SE}}(\widehat{\beta}_1) = \widehat{\sigma}\sqrt{\frac{1}{60} + \frac{1}{60}} = 4.34 \text{ m}$$

Including center makes the SE better calibrated: the residual variance now estimates the variation that actually contributes to the treatment comparison in this randomized block design.

PART 3

Modeling center as a random effect



From individual centers to a population of centers

- Besides estimating the treatment effect, the fixed-effects model estimates an effect for each of the six included centers.
- Often, our scientific interest is not in these particular centers or their individual effects.
- A random-intercept model accounts for center differences by describing how center effects vary across a population of centers.
- This gives a more parsimonious model and an estimate of the between-center variance.



The random-intercept model

RANDOM-INTERCEPT MODEL

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + c_j + \varepsilon_{ij}$$

$$c_j \overset{\text{iid}}{\sim} N(0, \sigma_c^2)$$

$$\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

For a given center effect, the expected walking distance is $\beta_0 + \beta_1 T_{ij} + c_j$.

Individual errors are independent of each other and of the center effects.

WHAT DOES THIS IMPLY FOR Y_{ij}?

Recap rules: slides 8–9

EXPECTED VALUE: ADD EXPECTATIONS

$$E(Y_{ij}) = \beta_0 + \beta_1 T_{ij} + E(c_j) + E(\varepsilon_{ij})$$
$$= \beta_0 + \beta_1 T_{ij}$$

VARIANCE: ADD INDEPENDENT VARIANCES

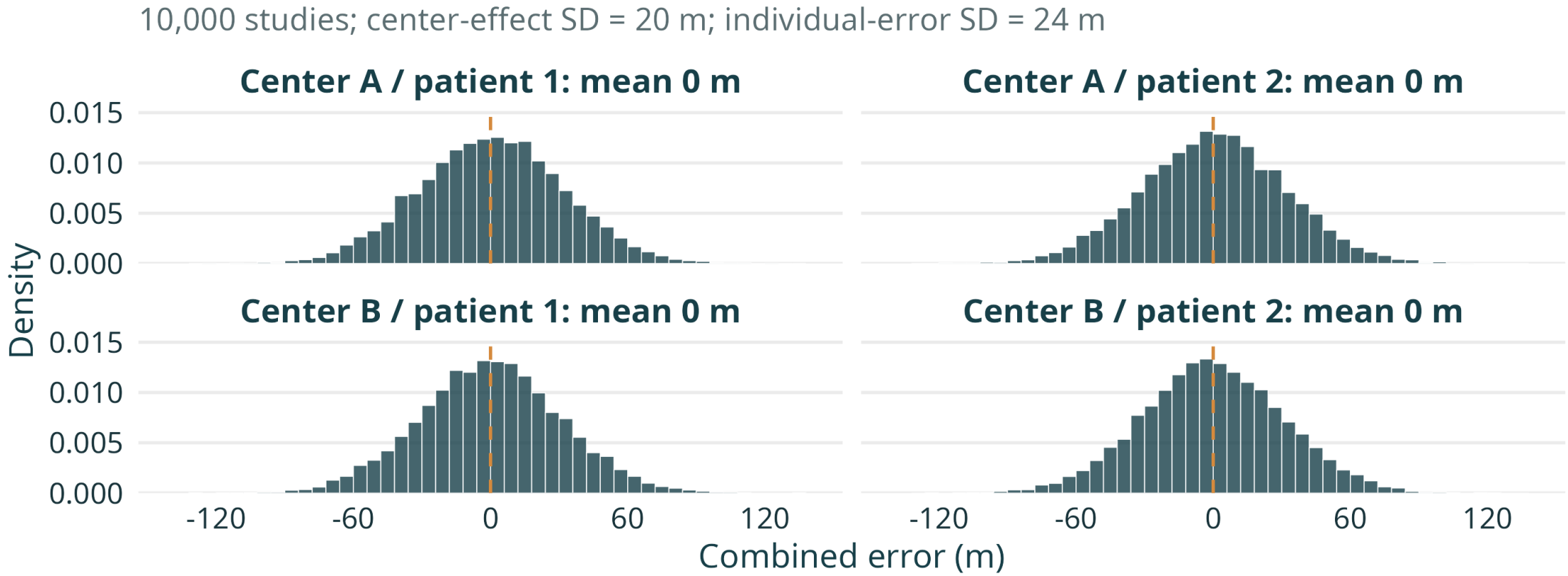
$$\text{Var}(Y_{ij}) = \text{Var}(c_j) + \text{Var}(\varepsilon_{ij})$$
$$= \sigma_c^2 + \sigma^2$$

Independent Normal terms give a Normal sum:

$$Y_{ij} \sim N\left(\beta_0 + \beta_1 T_{ij},\, \sigma_c^2 + \sigma^2\right)$$

Patients in the same center share the random effect c_j, so their outcomes are correlated.

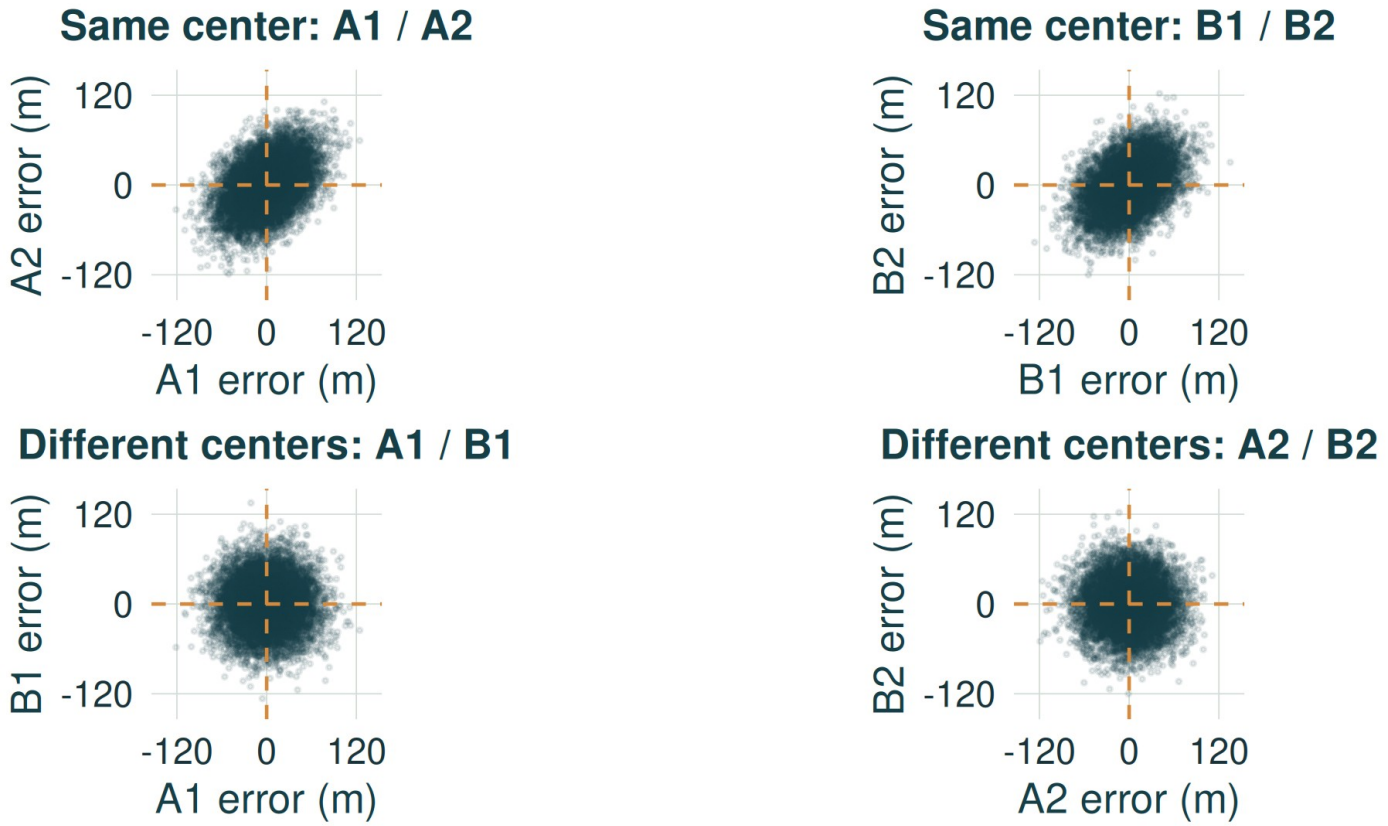
Combined errors: same mean and variance



Center effects and individual errors are drawn again in every simulated study. Dashed lines: mean 0.

Every combined error $c_j + \varepsilon_{ij}$ has mean 0 and variance $20^2 + 24^2 = 976$ m².

Combined errors: dependence within centers



Each dot pairs errors from one study; center effects and individual errors are redrawn in all 10,000 studies.

Same center: a shared c_j gives covariance $20^2 = 400$ m².
Different centers: independent combined errors, covariance 0.

Between-center variation and the ICC

After accounting for treatment

VARIANCE AND COVARIANCE

One patient:

$$\mathrm{Var}(Y_{ij}) = \sigma_c^2 + \sigma^2$$

Two different patients in the same center (i ≠ k):

$$\begin{aligned}\mathrm{Cov}(Y_{ij}, Y_{kj}) &= \mathrm{Var}(c_j) \\ &= \sigma_c^2\end{aligned}$$

Their individual errors are independent; their shared center effect creates the covariance.

FROM COVARIANCE TO CORRELATION

Divide their covariance by their common variance:

$$\mathrm{Corr}(Y_{ij}, Y_{kj}) = \frac{\sigma_c^2}{\sigma_c^2 + \sigma^2} = \mathrm{ICC}$$

The correlation between outcomes of two patients in the same center.

The proportion of remaining outcome variance attributable to differences between centers.

Time for lab 2



Comparing the models

Model	Estimate (m)	Standard error (m)
Treatment only	11.0	5.77
Fixed center effects	11.0	4.34
Random center effects	11.0	4.34

Balanced treatment allocation within centers makes the treatment estimates identical.

Both center models separate center variation from individual variation, giving better-calibrated standard errors in this example.

The treatment results agree here, but the models describe center variation differently: separate parameters for these six centers versus a distribution for a population of centers.

The End

Questions



